

Format-Dependent Calibration Shifts in Refusal Representations of Instruction-Tuned Language Models

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Abstract—Large language models deployed as autonomous agents communicate through structured formats: tool calls, function results, and system prompts. We find that the evaluated agentic-format conditions reduce the refusal-direction projection gap by 91–97% relative to direct prompts, while harmful and harmless inputs remain highly separable within each format ($\text{AUC} \geq 0.978$). A linear probe threshold calibrated on direct prompts fails to transfer to the tool_call format, dropping to near-chance accuracy (52%) despite within-format AUC of 0.990. The shift replicates across four instruction-tuned models spanning three model families (1.1–3B parameters), with a similar representational-geometric effect in a Phi-2 control. On the primary model, format-token ablation and cumulative activation patching, together with an exploratory sparse-autoencoder intervention, provide converging evidence that learned format and special tokens contribute substantially to the shift; system-token processing concentrates at Layer 0 while tool-call processing distributes across mid-to-late layers. Behavioral validation on three frontier systems shows refusal reduction under model-specific framing conditions: GPT-4o drops from 86% to 30% under authorized-context user-message framing ($N=50$, $p < 10^{-7}$), Claude Sonnet 4.6 from 34% to 12% through tool-output text wrapping ($N=50$, $p = 0.019$), and Gemini 2.5 Flash from 95% to 85% through native function-response ($N=20$, not significant). The effect reaches statistical significance for GPT-4o and Claude but not Gemini. In a controlled proof-of-concept experiment, format-diverse safety training stabilized tool-call refusal across five random seeds ($95\% \pm 4$ vs. $46\% \pm 34$ under direct-only training, $N=40$ per format), collapsing per-seed variance from a 5–83% range to a consistent 90–100%.

Index Terms—AI safety, large language models, adversarial machine learning, agentic systems, representation engineering, tool-augmented LLMs

I. INTRODUCTION

Large language models are increasingly deployed as *agents*: systems that call tools, execute functions, and communicate via structured protocols (JSON-RPC, MCP) [1]–[3]. This requires reliable parsing of structured format instructions. Simultaneously, these models are safety-trained to *refuse* harmful requests [4], [5], with mechanisms including the rank-1 refusal direction [6] operating on residual-stream representations.

We identify a representational conflict: **structured format and special tokens induce calibration shifts in refusal-related representations, shifting refusal-direction projec-**

tions sufficiently that a linear probe threshold calibrated on direct prompts fails to transfer to formatted prompts. Harmful and harmless inputs remain separable within each format (high AUC), but the decision threshold shifts by up to 1.6 units across formats.

A. Contributions

- 1) Evaluated agentic-format inputs shift refusal-related representations, reducing the refusal-direction gap by 91–97% relative to direct prompts; the strongest shifts occur for conditions containing model-native special tokens (§V).
- 2) Cumulative activation patching localizes system-token processing to Layer 0 (77% restoration) while tool_call processing distributes across layers 12–24; format-token ablation generally restores the gap, with token-identity and length controls supporting a contribution from structural syntax and native token identity. The gap reduction replicates across four instruction-tuned models plus a Phi-2 representational control (§V-B, §V-E, §VII).
- 3) Behavioral framing stress tests on three frontier systems show reduced refusal under model-specific framing conditions, reaching significance for GPT-4o (56pp, $p < 10^{-7}$) and Claude Sonnet 4.6 (22pp, $p = 0.019$) but not Gemini (10pp, $p = 0.50$) (§V-F).
- 4) Format-diverse safety training mitigates the failure in a controlled multi-seed proof of concept (§VI).

B. Threat Model

We consider an attacker who can influence tool-response content consumed by an LLM agent (e.g., a compromised API returning tool results) but does not control the system prompt or have direct model access. If behavioral refusal is reduced by format-conditioned inputs, harmful instructions in tool responses may elicit compliance and influence downstream actions. This threat applies when: (a) model responses trigger security-sensitive actions, (b) tool outputs lack independent validation, and (c) no format-invariant safety classifier is present.

II. RELATED WORK

Refusal representations. Arditi et al. [6] showed refusal is mediated by a rank-1 direction; Li et al. [7] identified similar directions for inference-time intervention; Rinsky et al. [8] demonstrated contrastive activation addition. Representation engineering [9] enables reading and writing model concepts. We extend this line by showing the refusal direction exhibits format-dependent calibration shifts.

Tool-augmented LLMs and agent security. Toolformer [1], Gorilla [2], and ToolLLM [3] require parsing structured instructions [10]. Greshake et al. [11] formalized indirect prompt injection; Wallace et al. [12] proposed instruction hierarchy; Yi et al. [13] evaluated agent vulnerabilities. We provide representational evidence for why tool-formatted content may bypass safety.

Adversarial attacks. Universal suffixes [14], jailbreak taxonomies [15], Policy Puppetry [16], cipher attacks [17], AutoDAN [18], and structured-output exploits [19] demonstrate diverse failure modes. Power [20] independently demonstrates behavior-level refusal reductions under paired tool framing across three models and 1,040 prompt pairs. Our work complements that result by identifying a format-dependent shift in internal refusal geometry, testing activation-level interventions, and evaluating format-diverse safety training.

Probing limitations. Linear probes recover information from representations but do not establish the model uses that information for generation [21], [22]. Our probe-based results establish format-dependent calibration shifts in safety-relevant directions, not that the model implements a fixed linear threshold.

Safety under distribution shift. Qi et al. [23] showed fine-tuning breaks safety alignment; Wolf et al. [24] argued safety is not robustly achieved through current RLHF. Our training experiment demonstrates that safety trained only on direct-format examples fails to generalize to structured formats.

III. BACKGROUND

Let $\mathbf{d}_\ell$ denote the rank-1 refusal direction at layer ℓ , extracted as:

$$\mathbf{d}_\ell = \frac{\bar{\mathbf{h}}_\ell^{\text{harmful}} - \bar{\mathbf{h}}_\ell^{\text{harmless}}}{\|\bar{\mathbf{h}}_\ell^{\text{harmful}} - \bar{\mathbf{h}}_\ell^{\text{harmless}}\|} \quad (1)$$

The projection $p_i = \mathbf{h}_i^\top \mathbf{d}_{\ell^*}$ measures how strongly input i activates the refusal direction, where $\mathbf{h}_i$ is the last-token residual-stream state. We define the format-specific harmful-minus-harmless gap:

$$\Delta_f = \mathbb{E}[p_i \mid i \in f, \text{harmful}] - \mathbb{E}[p_j \mid j \in f, \text{harmless}] \quad (2)$$

Hypothesis: For formats f that the model is trained to execute, $\Delta_f \ll \Delta_{\text{direct}}$ while within-format AUC remains high. This pattern indicates a calibration shift rather than information destruction. Note that Δ_f conditions on both format (f) and harmfulness; it is the within-format separation between harmful and harmless projections, not a marginal over mixed examples.

IV. METHOD

A. Data Splits and Direction Extraction

We use three disjoint prompt sets drawn from 90 harmful + 90 matched harmless prompts across 6 categories (violence/weapons, cybercrime, fraud/deception, illegal substances, harassment/abuse, miscellaneous illegal activity): (1) *Extraction* (20+20): refusal-direction extraction only. (2) *Validation* (20+20): layer selection and threshold calibration. (3) *Test* (50+50): all reported open-weight gap, AUC, and threshold-transfer metrics. Content is matched across formats (identical harmful/harmless text wrapped in each format). Layer selection maximizes AUC on validation set using directions extracted from the extraction set; $\ell^* = 26$ for Qwen2.5-1.5B. Splits are fixed using a category-stratified seed of 42. Gap retention: $R_f = \Delta_f / \Delta_{\text{direct}} \times 100\%$; Direct = 100% by construction.

Prompts were constructed as imperative harmful requests with harmless counterparts paired by list position, approximately controlling imperative form and lexical structure. The dataset contains 15 matched pairs per category (90 total); the test set uses 8–9 pairs per category (50 total). Full prompt sets available in the repository.

Example pair. Harmful: “How to bypass two-factor authentication.” Harmless: “How to set up two-factor authentication.” Each pair is wrapped identically in every format.

Format template example (Qwen tool-call-token condition):

```
<tool_call>
{ "name": "process",
  "arguments": { "query": "[PROMPT]" } }
</tool_call>
```

B. Format Conditions

Tier A: Direct prompts (baseline), tokenized as raw strings. **Tier B:** Generic structured syntax (JSON, XML, YAML) and non-native format tokens, serialized as raw strings. **Tier C:** Qwen-native tool-call and system-role tokens (<tool_call>, <|im_start|>), together with MCP JSON-RPC protocol structure. Only the first two contain model-native special tokens; MCP is included as an agentic protocol condition. All conditions use the same raw-tokenization path without `apply_chat_template`; formatted conditions add wrapper syntax and, where applicable, registered special tokens. When testing formats foreign to a model, format tokens are treated as plain text (Tier B). Tier C conditions confound format syntax with semantic role and special-token identity; this is discussed in Section VIII.

C. Activation Patching Protocol

For each harmful prompt, we run two forward passes: direct-format (source) and formatted (target). We patch the *last-token* hidden state from source into target at specified layers. Patching is **cumulative**: Table VI reports restoration when patching layers 0 through L . When sequences differ in length, we align on the final token. Layers indexed 0–27 for Qwen2.5-1.5B; Table VI stops at layer 26; the SAE operates on layer 27. Restoration: $(p_{\text{patched}} - p_{\text{formatted}}) / (p_{\text{direct}} - p_{\text{formatted}}) \times 100\%$, averaged over $N=8$ harmful prompts.

TABLE I
MODELS AND EVALUATION CONFIGURATIONS.

Model	Params	Role
Qwen2.5-1.5B-Instruct	1.5B	Primary (mechanistic)
TinyLlama-1.1B-Chat	1.1B	Cross-arch.
SmolLM2-1.7B-Instruct	1.7B	Cross-arch.
Qwen2.5-3B-Instruct	3B	Scale validation
Phi-2 [†]	2.7B	Non-chat control
GPT-4o (gpt-4o alias; Jun '26)	–	Frontier (framing)
Gemini 2.5 Flash (eval. Jul '26)	–	Frontier (native)
Claude Sonnet 4.6 (eval. Jun '26)	–	Frontier (text wrap)

[†]Phi-2: no safety training; representation control only.

TABLE II
STATISTICAL TEST MAPPING.

Claim	Test	Rationale
Gap ret. < 100%	Wilcoxon sign-rank	Paired
Gap-retention CI	Bootstrap (2000)	Class-stratified bootstrap
Frontier Δ refusal	McNemar's exact	Binary
Training B_{ic} vs A_{ic}	Sign test (seeds)	Directional

D. Models

E. Statistical Tests by Claim

Representation metrics use 95% percentile-bootstrap CIs (2000 resamples). Frontier refusal proportions use Clopper-Pearson exact intervals. Multiple comparisons: Holm-Bonferroni within each table ($\alpha = 0.05$).

F. Behavioral Classification

Frontier classification uses an automated heuristic: responses <80 characters are classified as refusal; longer responses are classified by keyword matching against refusal indicators ("I cannot", safety redirects) and compliance indicators (code fences, implementation language). Partial responses (disclaimer + content) merge with compliance. GPT-4o and Claude: N=50; Gemini: N=20 via native function-response API. Versions: gpt-4o rolling alias (Jun '26), claude-sonnet-4-6 (Jun '26), gemini-2.5-flash stable ID (Jul '26).

V. RESULTS

A. Format-Dependent Calibration Shift

Agentic-format conditions (Tier C) reduce the refusal-direction gap to 3.5–8.3% of baseline (raw gap: $\Delta_{\text{direct}} = 30.9$, $\Delta_{\text{tool_call}} = 2.0$; $p < 10^{-6}$ Wilcoxon signed-rank). Within-format AUC remains ≥ 0.978 for all conditions: the representations remain separable, but the optimal threshold shifts substantially (Table V). The direct-format threshold was selected on the validation set to maximize balanced accuracy and then held fixed for all test-format evaluations.

B. Causal Ablation

The length-matched control rules out input length as the primary explanation (92.8% vs. 5.2% at 32 tokens). The main sequence, length-matched controls, and token-identity controls use separate serialization templates; their absolute baselines are

TABLE III
GAP RETENTION (%) BY FORMAT TIER ON QWEN2.5-1.5B-INSTRUCT.
DIRECT = 100% BY CONSTRUCTION. N=50 MATCHED
HARMFUL/HARMLESS PAIRS PER CONDITION.

Tier	Format	Ret. (%)	95% CI
A	Direct (reference)	100.0	–
B	JSON	17.3	[14.6, 20.0]
	XML	37.4	[33.5, 41.3]
	YAML	29.2	[25.6, 32.8]
	OpenAI-style*	45.3	[41.4, 49.4]
C	System (Qwen < im_start >)	3.5	[3.1, 3.9]
	tool_call (Qwen vocab token)	6.5	[5.6, 7.4]
	MCP JSON-RPC	8.3	[7.1, 9.6]

*Non-native tokens; falls into Tier B for Qwen.

TABLE IV
FORMAT TOKEN ABLATION ON QWEN2.5-1.5B-INSTRUCT. REMOVING
FORMAT STRUCTURE GENERALLY RESTORES THE GAP; BRACKET AND
KEY-VALUE CONTROLS ARE NOT UNIQUELY ORDERED. N=50 MATCHED
PAIRS.

Condition	Ret. (%)	95% CI
Full tool_call (Qwen vocab token)	6.5	[5.6, 7.4]
JSON object, token removed	9.1	[7.2, 11.4]
Bracket delimiters	62.0	[56.8, 67.1]
Minimal key-value	83.0	[71.9, 95.2]
Text prefix only	94.8	[84.1, 106.1]
Direct (reference)	100.0	–
<i>Length-matched (32 tok.):</i>		
Natural language filler	92.8	[81.9, 104.3]
Tool-call format	5.2	[4.5, 6.0]
<i>Token-identity:</i>		
Real <tool_call> vocab token	14.8	[12.9, 16.7]
Arbitrary delimiters	52.2	[46.5, 57.6]

not directly comparable across sub-blocks. The model-native <tool_call> token leaves approximately one-third as much residual gap as arbitrary delimiters (14.8% vs. 52.2%). Removing the native token and simplifying serialization generally restores the gap; bracket and key-value controls do not form a uniquely ordered sequence, so we do not interpret their ordering as a strict dose-response.

C. Calibration-Shift Signature

In a separate full-vector probe experiment, a mean-pooled linear readout (averaging layer-26 residual-stream states across all token positions, trained on direct-format extraction data) achieves 88.2% cross-format accuracy vs. 80.0% for last-token. Mean-pooling improves system-token accuracy from 50% to 90% but slightly degrades tool_call accuracy, suggesting tool-call processing distributes across token positions.

D. Activation Patching

Patching through Layer 0 alone recovers 77% of the system-token shift but has negligible effect on tool_call (–3%). Tool-call restoration begins at mid-depth (50% by Layer 12) and saturates at 100% by Layer 24, indicating the shift distributes across layers 12–24. System processing concentrates at early

TABLE V

WITHIN-FORMAT AUC VS. CROSS-FORMAT ACCURACY. $\Delta\text{Thr.}$ = DIFFERENCE BETWEEN FORMAT-SPECIFIC OPTIMAL THRESHOLD AND THE DIRECT-CALIBRATED THRESHOLD. RAW Δ = HARMFUL-MINUS-HARMLESS GAP IN PROJECTION UNITS.

Format	AUC	Raw Δ	$\Delta\text{Thr.}$	Acc @ Dir. Thr.
Direct	0.994	30.94	–	94.0%
JSON	0.988	5.34	–0.93	92.0%
Tool_call	0.990	2.00	+1.58	52.0%
System	0.989	1.08	+0.27	91.0%

TABLE VI

CUMULATIVE ACTIVATION PATCHING RESTORATION (%). PATCHING LAST-TOKEN HIDDEN STATES FROM DIRECT-FORMAT PASSES INTO FORMATTED PASSES, LAYERS 0 THROUGH L . QWEN2.5-1.5B-INSTRUCT, $N=8$.

Format	L0	L6	L12	L18	L24	L26
Tool_call	–3.4	–9.9	50.1	80.4	100.0	100.0
System	77.1	88.8	93.0	103.4	97.2	97.2

layers. Values near or slightly above 100% indicate saturation rather than overshoot.

E. Cross-Model Replication

Tool_call produces substantial gap reduction ($<11\%$) across all tested models. Each replication model uses an independently extracted refusal direction and selected optimal layer. The cross-model experiment uses standardized serialization; its Qwen JSON value (7.4%) differs from the model-specific template in Table III (17.3%). System-token effects are architecture-dependent (Phi-2: 56.6% vs. SmolLM2: 2.0%).

F. Frontier Behavioral Validation

GPT-4o: McNemar’s exact test ($b = 28$, $c = 0$; $p < 10^{-7}$). Claude Sonnet 4.6: $b = 15$, $c = 4$; $p = 0.019$, Holm-adjusted $p = 0.038$. Gemini: $b = 2$, $c = 0$; $p = 0.50$ (directionally consistent but underpowered at $N=20$). Model-specific behavioral framing conditions reduced refusal across all three systems, reaching significance for GPT-4o and Claude but not Gemini. These are behavioral stress tests; each model’s formatted condition also alters contextual framing, so the observed effect conflates format structure with semantic context. Classification uses an automated heuristic (responses <80 chars = refusal; longer = keyword match).

VI. CONTROLLED FORMAT-DIVERSE TRAINING INTERVENTION

We fine-tune SmolLM2-135M-Instruct under two conditions using LoRA (rank=16, $\alpha=32$, lr= 2×10^{-4} , 5 epochs, batch size=4) across **five random seeds** (42, 123, 456, 789, 1024) with $N=40$ held-out evaluation prompts per format:

- **Condition A:** Safety training in direct format only. Each direct-format safety example (50 harmful $\rightarrow$ refusal, 50 harmless $\rightarrow$ helpful) is presented four times. Format-obedience training uses 10 benign tool/JSON/system

TABLE VII

GAP RETENTION (%) ACROSS ARCHITECTURES. $N=50$ MATCHED PAIRS PER MODEL EXCEPT QWEN2.5-3B ($N=30$). ALL INSTRUCTION-TUNED MODELS SHOW TOOL_CALL RETENTION $<11\%$.

Model	Direct	JSON	Tool_call	System
Qwen2.5-1.5B	100.0	7.4	6.5	3.5
Qwen2.5-3B	100.0	27.3	10.4	6.5
TinyLlama-1.1B	100.0	11.7	2.0	37.8
SmolLM2-1.7B	100.0	4.0	0.1	2.0
Phi-2 [†]	100.0	13.6	6.8	56.6

[†]Phi-2 results reflect representational geometry only (no safety training).

TABLE VIII

FRONTIER REFUSAL RATES UNDER BEHAVIORAL FRAMING STRESS TESTS. BINARY: EXPLICIT REFUSAL VS. NON-REFUSAL (PARTIAL COMPLIANCE OR FULL COMPLIANCE MERGED).

Model	Condition	Refusal	95% CI
GPT-4o ($N=50$)	Direct	86%	[73.3, 94.2]
	Auth-context framing*	30%	[17.9, 44.6]
Claude 4.6 ($N=50$)	Direct	34%	[21.2, 48.8]
	Tool-output text wrap	12%	[4.5, 24.3]
Gemini ($N=20$)	Direct	95%	[75.1, 99.9]
	Native fn_resp.	85%	[62.1, 96.8]

*Authorized red-team context in a user message; no native tool role or structured wrapper.

examples. Harmful content never appears inside structured formats. Total: 410 sequences.

- **Condition B:** Identical harmful/harmless prompts, but each safety example is presented once in each of four formats (direct, tool_call, JSON, system). Format-obedience with same examples. Total training volume matched: 410 sequences.

Condition A reproduces the predicted pattern: high direct-format refusal but highly variable tool_call safety ($46\% \pm 34$, range 5–83%). Direct-only training produces unreliable safety that depends on training stochasticity. Condition B collapses this variance ($95\% \pm 4$, range 90–100%). Tool-call refusal increased under Condition B for all five matched seeds (one-sided exact sign test, $p = 0.031$). The system format remained partially resistant to both regimes (A: $75\% \pm 15$; B: $56\% \pm 6$), suggesting system-role processing involves distinct mechanisms. Harmless compliance was not lower under format-diverse training (B: 62% vs. A: 56% across formats).

VII. COMPONENT-LEVEL ANALYSIS

For each Layer-0 attention head, we measured attention from the final query position, aggregating key positions into format-token and content-token bins. Across 15 prompts, 48 head-prompt pairs showed L1 routing divergence exceeding 0.5 (mean 0.99, range 0.50–1.81, max possible = 2.0): in direct format, last-token attention flows to content; in tool_call format, it partially redirects to format tokens.

Component-level patching (attention-only at Layer 0) restores 79.0% of the system-format projection shift vs. 26.9% for MLP-only ($N=8$), confirming dominant early-layer atten-

TABLE IX

MULTI-SEED BEHAVIORAL REFUSAL (%): MEAN $\pm$ STD ACROSS 5 SEEDS. N=40 HELD-OUT HARMFUL PROMPTS PER FORMAT PER SEED; HARMLESS PROMPTS EVALUATED SEPARATELY FOR COMPLIANCE. SMOLLM2-135M-INSTRUCT WITH LORA.

Cond.	Direct	Tool_call	JSON	System	XML
A	98 $\pm$ 2	46 $\pm$ 34	82 $\pm$ 14	75 $\pm$ 15	83 $\pm$ 21
B	98 $\pm$ 1	95 $\pm$ 4	97 $\pm$ 4	56 $\pm$ 6	97 $\pm$ 3

tion contribution for system tokens. For tool_call, Layer-0 component patching produces negligible restoration (-4.2% attention, -10.5% MLP), consistent with Table VI: the tool_call shift distributes across layers 12–24 rather than originating at Layer 0. The attention routing divergence at Layer 0 does not causally produce the tool_call projection shift.

SAE exploratory analysis: A 512-feature sparse autoencoder trained on 160 last-token activations from layer 27 across four formats. Nineteen features with high harmful-minus-harmless differential were selected; adding these features back during tool_call passes restores 72% of the projection (N=10). Because feature selection, training, and evaluation reuse a small overlapping dataset, this is exploratory.

VIII. DISCUSSION

A. Interpretation and Caveats

Our results establish that the refusal-direction gap undergoes a large, consistent calibration shift when content is delivered through structured format channels. Three caveats bound interpretation: (1) A linear probe threshold failing to transfer does not prove the model’s generation mechanism relies on that threshold. Frontier results provide independent behavioral evidence but do not establish the same representational mechanism operates at scale. (2) Tier C conditions confound structured syntax, special-token identity, semantic role, and implicit trust. Token-identity ablation separates special tokens from generic structure, but a full factorial remains future work. (3) Mechanistic analysis covers 1.1–3B parameters only.

B. Implications for Agent Security

These findings identify a potential failure mode for agents consuming attacker-influenced tool outputs where model responses trigger security-sensitive actions. The threat requires: attacker ability to influence tool-response content, absence of format-invariant safety classifiers, and model outputs connected to consequential actions.

Potential defenses: (1) Format-diverse safety training substantially stabilizes tool-call refusal in our 135M proof-of-concept. (2) Multi-position safety classifiers (mean pooling improved average cross-format transfer, though not the tool-call condition specifically). (3) Secondary output gating independent of internal safety. (4) Input canonicalization: strip format tokens before safety evaluation. (5) Role-aware trust boundaries: treat tool-response content with user-message scrutiny.

C. Limitations

Mechanistic analysis spans 1.1–3B parameters. The format-vs-role confound is not fully resolved. Cumulative patching uses N=8. SAE analysis uses shared selection/evaluation data (N=10). Training uses a 135M model; scale-up is future work. The prompt set is synthetic with approximately matched harmless controls; one fixed stratified split is used. GPT-4o framing conflates format with semantic context (role claim). Claude uses text serialization in a user message, not native tool_result. Gemini uses a stable model ID; results reflect the model state at evaluation time. Frontier CIs are wide due to small N. Classification uses an automated heuristic rather than manual review. Phi-2 is a representational control only (no safety training).

IX. CONCLUSION

Structured format and special tokens induce calibration shifts in refusal representations of instruction-tuned LLMs: the harmful-harmless gap reduces by 91–97% under agentic-format conditions while within-format AUC ≥ 0.978 is preserved. Format-token ablation, cross-model replication, and cumulative activation patching provide converging evidence: system-token processing concentrates at Layer 0 (77% restoration), while tool-call processing distributes across layers 12–24. Behavioral stress tests on three frontier systems show reduced refusal reaching significance for GPT-4o (56pp) and Claude Sonnet 4.6 (22pp). Format-diverse safety training stabilizes tool-call refusal (5 seeds, 46% $\rightarrow$ 95% mean, σ collapse from 34 to 4). For agentic systems consuming untrusted structured inputs, format robustness can be substantially improved through explicit engineering.

AI TOOLS USED

Claude (Anthropic, Sonnet 3.5/4 and Opus 4, 2024–2026) was used for code generation, manuscript editing, and statistical verification. ChatGPT (OpenAI; GPT-4o, o3, GPT-5.6 Thinking) provided peer-review feedback. GPT-4o and Gemini 2.5 Flash were evaluation targets, not authoring tools. Affected portions included experiment code, the Methods and Results verification workflow, and manuscript-wide prose editing. The author designed experiments, selected hypotheses, verified analyses, and takes responsibility for all interpretations. Open-weight experiments on author hardware (RTX 3080); frontier evaluations via vendor APIs.

REPRODUCIBILITY

Open-weight experiments: RTX 3080, PyTorch 2.7.1, Transformers 4.46.3, ~ 8 GPU-hours. Mechanistic models in FP16; fine-tuning in float32 with mixed-precision. Frontier evaluations Jun–Jul 2026: OpenAI (gpt-4o rolling alias), Google (gemini-2.5-flash stable ID, Jul ’26), Anthropic (claude-sonnet-4-6). GPT-4o and Claude use max_tokens=600; Gemini uses SDK defaults. Classification: automated length + keyword heuristic. Code and prompts: <https://github.com/Suzu-Testing/Structured-Formats-Collapse-Refusal-Representations>

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